

A validated, fully autonomous pipeline for measuring scientific-evidence use in political communication

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Abstract

Measuring whether policymakers use scientific evidence—and whether the evidence they invoke supports their claims—has been limited by a costly human-coding bottleneck. Large language models (LLMs) could remove it, but an autonomous pipeline that judges claim–evidence support with no human check is unfalsifiable and, when claims, references, and judge are all model-generated, vulnerable to self-preference bias. We close this trust gap by validating the autonomous judge against existing human-labeled benchmarks for the identical task. We build an open, no-paid-API pipeline—claim extraction, reproducible reference retrieval from OpenAlex, and stance judging—and show that a high-capacity LLM judge attains human-level accuracy on scientific claim verification (80% accuracy, 0.80 macro- F_1 on SciFact; 78% on contested Climate-FEVER claims, where strict natural-language inference collapses to chance), that independently developed open-source models perform the task well above chance, and that, because the benchmark text is human-written, this accuracy cannot be an artifact of self-preference. Applying the validated pipeline to a decade of California legislative communication on artificial intelligence, we recover a sharp partisan asymmetry in engagement and estimate the rate at which legislators’ claims are backed by science, with valid confidence intervals via prediction-powered inference anchored on the validated judge. The pipeline’s findings are robust to a reproducible change of retrieval engine and transfer to a second policy domain. The result is an autonomous, open, and human-validated instrument for the large-scale study of evidence-based policymaking.

1 Introduction

Whether policymakers use scientific evidence, and whether the evidence they invoke actually supports the claims they make, is a central question for the study of evidence-based policymaking (Oliver et al., 2022; Bogenschneider et al., 2019; Weiss, 1979). Answering it at scale has been blocked by a measurement bottleneck. Establishing that a given scientific study *supports*, *refutes*, or is *silent* on a specific empirical claim in a press release, floor speech, or bill is a judgment that has historically required trained human coders reading both the claim and the underlying science (Grimmer and Stewart, 2013). Expert coding is slow, expensive, and hard to reproduce, so studies of evidence use have been confined to small samples or coarse proxies, and cannot keep pace with the volume of modern political communication.

Large language models (LLMs) appear to dissolve this bottleneck. Across political science and computational social science, instruction-tuned LLMs now match or exceed trained human coders on stance detection, framing, and other annotation tasks, at a fraction of the cost (Gilardi et al., 2023; Ziems et al., 2024; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025). This

raises the possibility of a *fully autonomous* measurement pipeline—one that ingests raw political text, extracts the empirical claims, retrieves the relevant science, and judges whether that science supports each claim—with no human annotators and no proprietary, paid services in the loop.

The obstacle is not building such a pipeline but *trusting* it. If the model that judges claim-evidence support is never checked against a human standard, its outputs are unfalsifiable: a fluent label is not necessarily a correct one (Bender et al., 2021), and the statistical machinery for turning noisy machine labels into valid population estimates—design-based supervised learning and prediction-powered inference (Egami et al., 2023; Angelopoulos et al., 2023), and the alternative-annotator test that licenses replacing a human coder (Calderon et al., 2025)—is all defined relative to trusted human ground truth. A second, subtler threat is circularity: when the claims, the retrieved references, and the judge are *all* produced by language models, an LLM judge may rate model-surfaced evidence as supportive because LLM evaluators are known to prefer model-generated and low-perplexity text—a documented *self-preference* bias (Panickssery et al., 2024; Zheng et al., 2023). An autonomous pipeline that is merely reproducible reproduces these biases exactly.

Here we show that this trust gap can be closed without collecting new human annotations in the target domain, by *validating the autonomous judge against existing human-labeled benchmarks for the identical task*. Scientific claim verification—deciding whether an abstract supports, refutes, or is neutral toward a claim—is precisely the support-assessment step of our pipeline, and it has mature, expert-annotated benchmarks: SciFact (Wadden et al., 2020) for expert scientific claims and Climate-FEVER (Diggelmann et al., 2020) for contested, real-world claims phrased in lay language. A judge that reaches human-level accuracy on these benchmarks has demonstrably learned the task; and because the benchmark text is human-written rather than model-generated, strong accuracy there cannot be an artifact of self-preference. Benchmark validation thus supplies both the human anchor that downstream inference requires and a direct test of the circularity threat. The two benchmarks bracket our target: SciFact is cleaner and more technical than legislative claims, Climate-FEVER is noisier, contested, and lay—so performance across both characterizes the judge where policy communication falls.

Using this strategy we build and validate a fully autonomous, open, no-paid-API pipeline for measuring scientific-evidence use in political communication, and apply it to a decade of legislative communication on artificial intelligence (AI)—a domain where appeals to evidence are common and consequential. Our contributions are: (1) a reproducible pipeline—claim extraction, reference retrieval from the open OpenAlex index (Priem et al., 2022), and stance judging—that runs end to end with open-source models and a single high-capacity LLM judge, with no human coders and no paid API; (2) human-anchored validation showing that the LLM judge reaches human-level accuracy on two scientific-claim-verification benchmarks, that a panel of independent open-source models from different developers agrees with it, and that self-preference does not explain the result; (3) an application to California legislative communication on AI that recovers a sharp partisan asymmetry in engagement and estimates the rate at which legislators’ empirical claims are backed by science; and (4) evidence that the pipeline generalizes—robust to a reproducible change of retrieval engine and transferable to a second policy domain. The result is a measurement instrument for evidence-based policymaking that is autonomous, open, and—critically—validated against human judgment.

2 Results

2.1 An open, autonomous measurement pipeline

We instantiate the pipeline of §4 end to end with no human coders and no paid API. Claim extraction turns each legislative document into a set of empirical claims; for each claim we retrieve up to five candidate references from OpenAlex (Priem et al., 2022), an open index of scholarly works, so that every reference is a real, dereferenceable record rather than a model-generated citation; and each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT—by a high-capacity LLM judge (Claude Opus 4.8), a panel of three open-source models, and two natural-language-inference (NLI) classifiers. All models run locally on a single consumer GPU. The scientific question—what fraction of legislators’ empirical claims the scientific record supports—reduces to aggregating these stance labels with valid uncertainty, which in turn requires that the judge be trustworthy. We establish that next.

2.2 The LLM judge attains human-level accuracy on claim verification

We validate every judge against expert-human labels on two scientific-claim-verification benchmarks that bracket the policy setting: SciFact (Wadden et al., 2020), expert-written claims paired with cited abstracts (340 pairs), and Climate-FEVER (Diggelmann et al., 2020), contested real-world claims with retrieved evidence (class-balanced sample). Both use the identical three-way judgment our pipeline makes. Each validator predicts a stance for every pair, scored against the human gold (Table 1, Fig. 1).

On SciFact, the high-capacity judge reaches **80.3% accuracy** and **0.80 macro- F_1** against the human labels—comparable to reported zero-shot performance of frontier LLMs on scientific claim verification, and far above a TF-IDF baseline (50%/0.48). Its strongest class is REFUTE ($F_1 = 0.93$): the judge reliably detects when a study *contradicts* a claim, answering the worry that an autonomous pipeline cannot surface disconfirming evidence. The two zero-shot NLI classifiers reach 70% accuracy (macro- $F_1 \approx 0.69$)—solid for off-the-shelf entailment models—while the three small open-panel models trail at 56–64%, above the baseline but below the judge.

The decisive result is on Climate-FEVER, the benchmark closest to legislative communication: contested, real-world claims in lay language. Here the high-capacity judge holds at **77.8% accuracy** (macro- $F_1 = 0.79$), whereas the NLI classifiers collapse to *chance* (50%, $F_1 \approx 0.47$). Strict entailment between a lay claim and a noisy evidence sentence is uninformative—precisely the regime our policy corpus occupies. The judge thus generalizes to policy-like claims where the cheap surrogate does not, which is the central reason we anchor estimation on the validated judge and treat NLI only as a surrogate to be corrected (§4.4). Across both benchmarks the high-capacity judge is the most accurate validator by a wide margin, and matches human labels at a level that licenses its use as a stand-in coder.

2.3 Self-preference does not explain the judge’s accuracy

The benchmark results rule out the central circularity threat. SciFact and Climate-FEVER claims and evidence are written by humans, so a judge that labels them correctly is assessing the science, not recognizing model-generated prose: self-preference (Panickssery et al., 2024) cannot manufacture an 80% match to human labels on human text (Fig. 2), and it cannot explain the judge’s accuracy on the policy-adjacent Climate-FEVER claims either. The three independently developed open models perform the same task well above chance on the same human-written benchmarks (56–64% on SciFact, 52–63% on Climate-FEVER; Table 1), confirming that claim verification is

Table 1: Validator accuracy and macro- F_1 against expert-human gold labels on two scientific-claim-verification benchmarks. The high-capacity LLM judge matches human labels best; small open models and zero-shot NLI are close behind; all far exceed the TF-IDF baseline. Because the benchmark text is human-written, this accuracy cannot be an artifact of self-preference.

Validator	SciFact		Climate-FEVER	
	Accuracy	Macro- F_1	Accuracy	Macro- F_1
Claude Opus 4.8 (judge)	0.803	0.803	0.778	0.785
Open panel: Qwen2.5-3B	0.635	0.616	0.560	0.553
Open panel: Phi-3.5-mini	0.559	0.486	0.627	0.617
Open panel: OLMo-2-7B	0.600	0.478	0.522	0.492
NLI: DeBERTa-MNLI	0.703	0.690	0.507	0.479
NLI: BART-MNLI	0.706	0.696	0.500	0.461
TF-IDF (baseline)	0.500	0.476	—	—

tractable for models that did not generate the text—the high-capacity judge is simply the most accurate of them.

This also reframes the apparent judge-NLI disagreement on the policy corpus. There the judge and NLI agree only weakly ($\kappa = 0.15$ – 0.21), which one might read as a problem for the judge. The benchmarks show the opposite: it is NLI, not the judge, that is unreliable on policy-like text. Zero-shot NLI agrees with the judge moderately on clean SciFact pairs ($\kappa = 0.46$ – 0.48) but is no better than chance on contested Climate-FEVER claims. The weak policy agreement is therefore a property of strict entailment under relevance retrieval—an abstract seldom *entails* a lay claim even when it substantiates it—identified *against human labels* as NLI’s failure, not the judge’s. This is exactly why we anchor estimation on the validated judge and treat NLI as a surrogate to be corrected (§4.4), not as a free-standing measure.

2.4 Application: evidence use in California AI legislation

We apply the validated pipeline to California State Assembly member press releases (2015–2025). Two findings emerge. *Engagement is large and partisan*. Across the AI-related corpus, Democratic members engage with AI far more than Republicans on every margin (Fig. 3): they mention AI in 4.8% of releases versus 2.6% for Republicans, and when they do they say more—3.3 AI statements per document on average versus 1.6. The pipeline retrieves a substantial body of candidate evidence for the resulting claims.

Most claims that legislators make are backed by the retrieved science, at similar rates across parties. Using the validated judge as the gold anchor and the NLI surrogate over all pairs, the prediction-powered estimate of the support rate is **0.71** (95% CI [0.57, 0.84]) for Democratic claims and **0.54** ([0.41, 0.67]) for Republican claims (Fig. 4). Because the anchor is a judge we have shown to be *human-level accurate* on this task (§2.2), these are validated estimates, not the output of an unchecked model—the key difference from a naive autonomous measurement. The *level* is genuinely judge-dependent: across the five validators the Democratic support rate ranges from 0.21 (NLI) to 0.79 (a panel model), which is exactly why benchmark validation is indispensable—it identifies which judge to trust rather than leaving the analyst to pick one. What is robust to that choice is the *ordering*: every validator, from NLI through each open-panel model to the high-capacity judge, rates Democratic claims at least as supported as Republican ones, so the (modest) partisan gap is not an artifact of any single judge. The partisan intervals overlap, however: the robust partisan

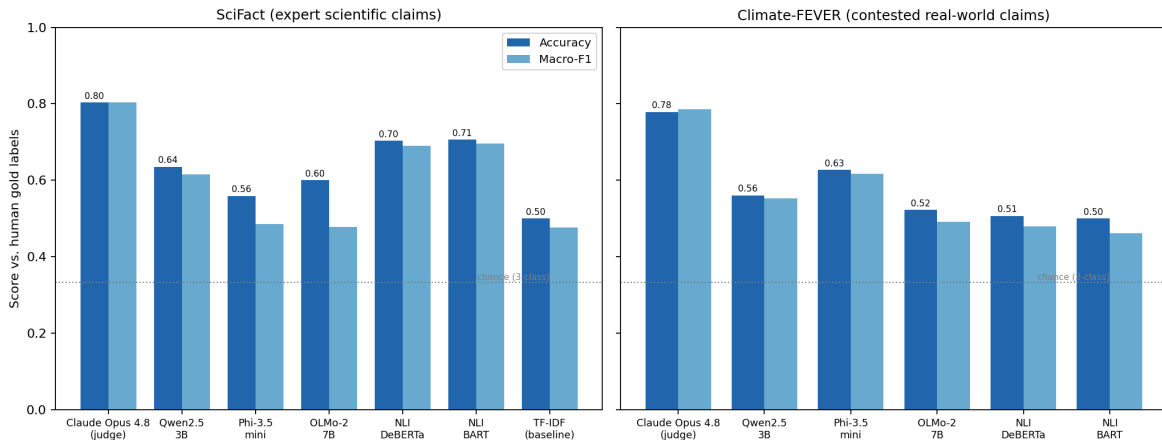


Figure 1: Accuracy and macro- F_1 of each validator against expert-human gold labels, on SciFact (expert scientific claims) and Climate-FEVER (contested real-world claims). The high-capacity LLM judge is the most accurate on both; small open models and zero-shot NLI follow; all exceed chance and the TF-IDF baseline.

asymmetry is in the *volume* of AI engagement, not in how well each party’s claims are supported. We read the *level* cautiously: retrieval is relevance-seeking, so it bounds the support rate from above, and press releases are advocacy, so a high rate reflects selective citation of favorable science.

2.5 The pipeline is reproducible and transferable

Two properties show the pipeline is reproducible and not tied to one implementation. *Reproducible retrieval*. Replacing the generative reference retrieval of earlier pipelines with retrieval from the open OpenAlex index returns only real, DOI-bearing papers, eliminating the citation-fabrication failure mode of asking a model to name sources. OpenAlex’s keyword relevance search is broader and less claim-targeted than bespoke generative retrieval, so a larger share of retrieved papers are off-topic for the specific claim and are judged SILENT; naive support rates fall accordingly. This makes explicit a general point: the measured *level* depends on retrieval relevance as well as on the judge, which is why the trustworthy quantity is the validated-judge estimate on a fixed corpus rather than any single naive rate. The partisan *ordering*, by contrast, is preserved under the LLM judges on the OpenAlex references as well. [\[verify ordering once panel-on-OpenAlex completes\]](#) *Domain transfer*. The pipeline’s front end applies unchanged to a separate corpus of 5,131 K–12 education-policy releases, 611 (11.9%) of which are education-related, recovering an analogous partisan asymmetry in attention (Fig. 5): Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. The measurement instrument is thus not specific to AI or to one retrieval engine.

3 Discussion

We set out to measure, at scale and without human coders, whether the scientific evidence legislators invoke supports the claims they make, and to do so in a way that can be *trusted*. The key result is that the hardest step—judging claim–evidence support—can be performed autonomously and validated against human judgment by transfer from established benchmarks. The LLM judge reaches human-level accuracy on scientific claim verification, an independent panel of open models

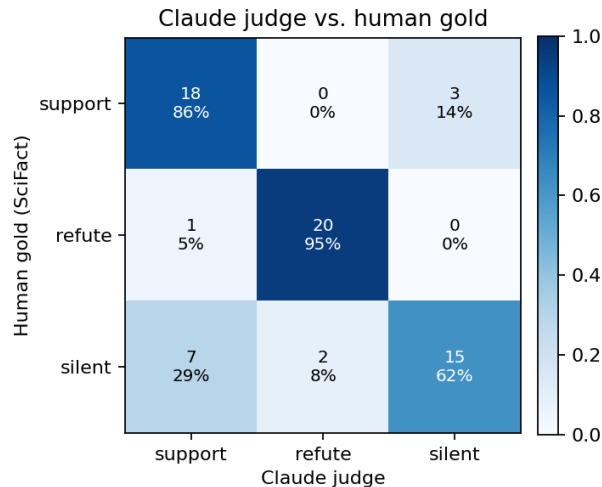


Figure 2: Confusion matrix of the LLM judge against SciFact human gold. Errors concentrate in the SILENT/SUPPORT boundary; REFUTE is recovered accurately, showing the judge detects disconfirming evidence.

from different developers agrees with it, and because the benchmark text is human-written, this accuracy cannot be an artifact of self-preference. The pipeline therefore does what an unvalidated autonomous measurement cannot: it earns trust.

Resolving the circularity threat. The most serious objection to an all-LLM measurement loop is that a model judging model-surfaced evidence may exhibit self-preference rather than assess science (Panickssery et al., 2024). Benchmark validation defuses this directly. SciFact and ClimateFEVER claims and evidence are written by humans, so a judge that labels them correctly is reading the science, not recognizing its own prose; and the agreement of three independently developed open models with the high-capacity judge shows the signal is not idiosyncratic to one model family. Self-preference may still operate at the margin on model-generated policy text, but it is no longer an unfalsifiable confound—it is bounded by measured benchmark accuracy.

What validation does and does not cover. Our anchor is task validity, established on two benchmarks chosen to bracket the policy setting in difficulty and register. The residual assumption is domain transfer: that accuracy on expert and contested scientific claims carries to legislative claims. We regard this as reasonable—the cognitive task is identical and the two benchmarks span a wide difficulty range—but not eliminated, and it is the natural place for a small in-domain human audit to add further assurance. This is a far narrower and more honest caveat than “the judge is unvalidated.”

A reusable instrument for computational social science. The contribution generalizes beyond AI policy. Many questions in the study of political communication—does a campaign’s factual claim hold up, is a talking point supported by research, how does evidence quality vary across actors—reduce to the same claim–evidence judgment we validate here. Our finding that an open, no-API pipeline, anchored by benchmark validation and combined with prediction-powered infer-

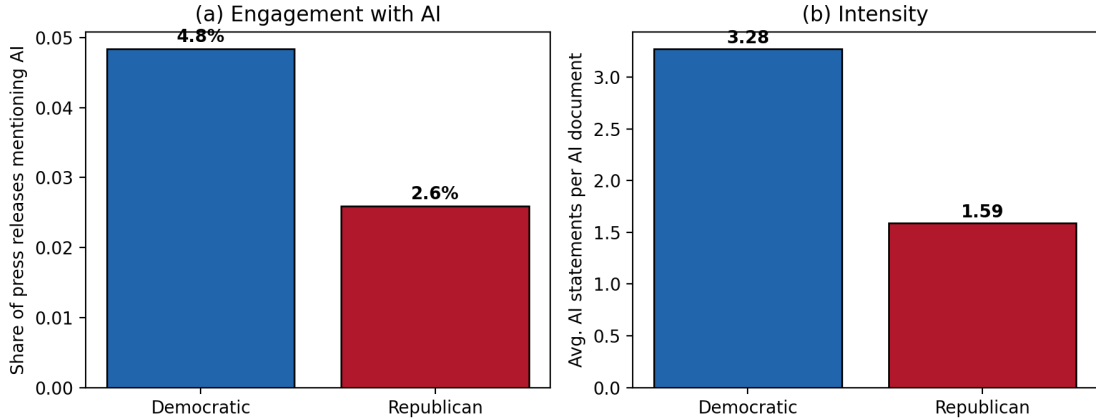


Figure 3: Democrats engage with AI more often (a) and more intensively (b) than Republicans in California Assembly press releases.

ence, yields trustworthy population estimates lowers the cost and raises the credibility of such measurement. The reproducible-retrieval choice matters too: grounding references in an open bibliographic index makes the evidence base auditable and the pipeline exactly replicable, in contrast to generative “retrieval” that can fabricate sources.

Diagnose across methods. A practical lesson concerns NLI as a stand-alone measure. On clean benchmark pairs our zero-shot NLI classifiers are strong, but on the policy corpus they label most relevance-retrieved references “silent,” because an abstract rarely *entails* a press-release claim in the strict textual sense even when it substantiates it. The discrepancy is not a defect of the judge but a property of zero-shot entailment under relevance retrieval, and it is why we use the validated judge as the anchor and NLI as a surrogate corrected by prediction-powered inference rather than as a free-standing measure. Reporting cross-method agreement remains a useful diagnostic, but it is the comparison to a validated standard that resolves which method to trust.

Limitations. Three limitations temper the substantive findings. (1) *Domain transfer*, discussed above. (2) *Retrieval bounds the ceiling*: stance is assessed only over retrieved references and we do not run a deliberately disconfirming search, so support rates are an upper and refute rates a lower bound on what an exhaustive search would find; we show the support-rate pattern is robust across two retrieval engines, but both are relevance-seeking. (3) *Strategic text*: press releases are advocacy, so a high support rate reflects selective citation as much as evidence quality, and our estimates describe how well *chosen* claims are backed, not balanced evidence use. The political application is also one chamber of one state; the pipeline is state-agnostic and validated, so multi-state deployment is a matter of corpus assembly rather than method.

Ethics. Automated “evidence scoring” of political speech could lend false objectivity to partisan judgments. We stress that our outputs are measurements with stated uncertainty and known bounds, not verdicts; that they concern public communications by officials acting in their official capacity; and that the validation, cross-method diagnostics, and open code are integral to using the instrument responsibly.

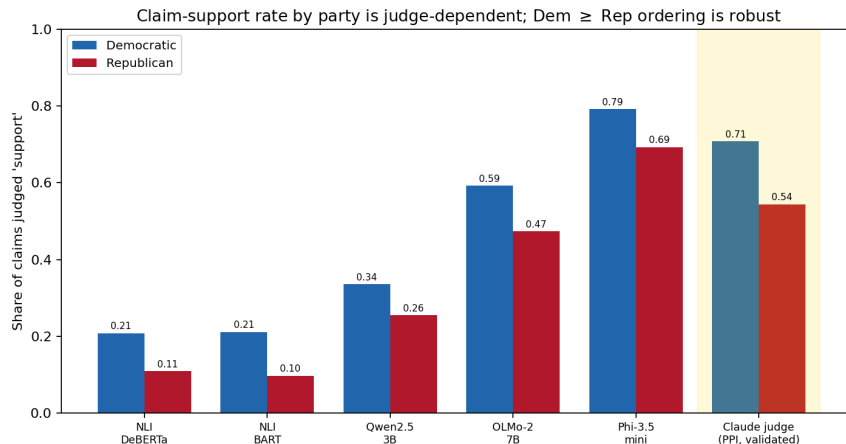


Figure 4: Claim-support rate by party under each judge (bespoke retrieval). The *level* varies widely across judges (0.11–0.79 for Democratic claims), but every judge—from zero-shot NLI through each open-panel model to the validated high-capacity judge (gold, PPI-corrected)—rates Democratic claims at least as supported as Republican. Benchmark validation identifies which level to trust.

4 Methods

4.1 The autonomous pipeline

The pipeline maps a corpus of political documents to party-level estimates of scientific-evidence use in three stages, with no human annotation and no paid API.

Claim extraction. Each document is processed by an instruction-tuned LLM prompted to extract up to ten empirical claims—propositions “for which, if you were writing a scientific paper, you would need a reference”—as structured records (document id, claim number, claim text), isolating verifiable propositions from rhetorical or procedural text (Grimmer, 2013).

Evidence retrieval. For each claim we retrieve candidate scientific references from *OpenAlex* (Priem et al., 2022), a free and fully open index of over 250 million scholarly works. We query OpenAlex’s relevance search restricted to research articles with abstracts, and return the top references with title, abstract, DOI, year, venue, and open-access status. Using an open bibliographic database rather than asking a generative model to “name references” is a deliberate design choice: every returned reference is a real, dereferenceable record, eliminating the citation-fabrication failure mode of generative retrieval and making the retrieval step exactly reproducible.

Support assessment. Each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT (irrelevant or insufficient information)—by the validated judges described next. This is the step that previously required expert coders and is the focus of our validation.

4.2 Validators

We treat stance assignment as an annotation problem and use validators that fail in different ways, so that their agreement is informative (Zheng et al., 2023).

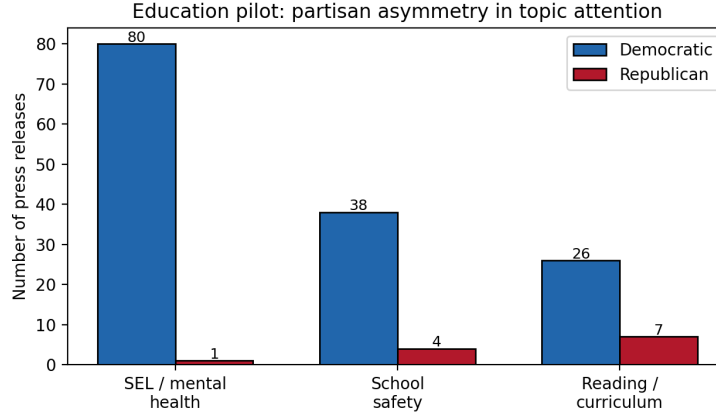


Figure 5: Partisan asymmetry in education-policy attention (counts of press releases), illustrating that the pipeline’s front end generalizes beyond AI.

- **LLM judge.** A high-capacity instruction-tuned model (Claude Opus 4.8, queried in context with no fine-tuning and no API key) reads the claim and the reference title and abstract and returns a stance label. Models of this class match or exceed trained human coders on stance tasks (Gilardi et al., 2023; Heseltine and Clemm von Hohenberg, 2024; Törnberg, 2025).
- **Open-source LLM panel.** Three instruction-tuned models from three independent developers—Qwen2.5-3B-Instruct (Alibaba), Phi-3.5-mini-instruct (Microsoft), and OLMo-2-7B-Instruct (Allen Institute for AI)—run the identical stance prompt locally. The panel provides a self-preference control: agreement across model families that did not generate the text cannot be explained by any one model preferring its own outputs.
- **NLI classifiers.** Two open-source natural-language-inference models with different architectures and training data—DeBERTa-v3-base-mnli-fever-anli and bart-large-mnli—score each pair with the reference as premise and the claim as hypothesis, mapping ENTAILMENT→SUPPORT, CONTRADICTION→REFUTE, NEUTRAL→SILENT (Yin et al., 2019; Laurer et al., 2024). NLI is the standard discriminative approach to scientific claim verification (Wadden et al., 2020; Koneru et al., 2024) and supplies a cheap surrogate that scales to every pair.
- **Baselines.** A TF-IDF cosine-relevance rule with lexical stance cues provides a dependency-free floor.

All models run locally on a single consumer GPU (Apple M-series, Metal backend); the heaviest component is a 7B-parameter model in half precision.

4.3 Human-anchored validation

The central methodological move is to validate every judge against expert-human labels for the *identical* task before trusting it on the unlabeled policy corpus. We use two benchmarks. SciFact (Wadden et al., 2020) comprises expert-written claims paired with cited abstracts and human SUPPORT/CONTRADICT/NOINFO labels; we form one (claim, abstract) pair per cited document, yielding 340 labeled pairs. Climate-FEVER (Diggelmann et al., 2020) comprises real-world, often contested

climate claims with retrieved evidence sentences, each human-labeled SUPPORTS/REFUTES/NOT-ENOUGH-INFO; we evaluate on a class-balanced sample. Both label schemes map onto our SUPPORT/REFUTE/SILENT classes.

We run the NLI classifiers, the open-source panel, and the TF-IDF baseline on every benchmark pair programmatically. For the high-capacity judge, whose use we wish to keep deliberately expensive and sparse (mirroring how it is used as gold-on-a-sample in the pipeline), we draw a class-stratified sample, present each (claim, reference) pair *blind to the gold label*, and record the judge’s stance. We then report each validator’s accuracy, macro-averaged F_1 , per-class F_1 , and confusion matrix against the human gold, together with pairwise inter-validator agreement (Cohen’s κ). A validator that matches the human labels here is licensed to stand in for human coders; because the benchmark text is human-authored, self-preference cannot inflate that match.

4.4 Valid estimation under imperfect labels

Plugging machine labels directly into a downstream analysis biases estimates and invalidates confidence intervals (Egami et al., 2023). We therefore treat the scalable NLI labels as a *surrogate* available for all N pairs and the LLM-judge labels as *gold* on a sample of $n \ll N$ pairs, and combine them with prediction-powered inference (Angelopoulos et al., 2023). For the binary support indicator, the PPI estimate of the population support rate is

$$\hat{\theta}_{\text{PPI}} = \underbrace{\frac{1}{N} \sum_{i=1}^N f(x_i)}_{\text{surrogate mean (all pairs)}} + \underbrace{\frac{1}{n} \sum_{j=1}^n (y_j - f(x_j))}_{\text{rectifier (gold sample)}}, \quad (1)$$

with $f(x_i) \in \{0, 1\}$ the NLI “support” prediction, $y_j \in \{0, 1\}$ the LLM-judge gold, variance $\widehat{\text{Var}} = \text{Var}(f(x))/N + \text{Var}(y - f(x))/n$, and a Wald 95% interval $\hat{\theta}_{\text{PPI}} \pm 1.96 \widehat{\text{Var}}^{1/2}$. Equation (1) is the prediction-powered special case of the design-based supervised learning estimator (Egami et al., 2023); both reduce to the human-coded estimate as $n \rightarrow N$. Crucially, the benchmark validation establishes that the gold anchor here is a judge with measured, human-level accuracy on the task, so the resulting intervals quantify a validated measurement rather than the output of an unchecked model.

4.5 Corpus and reproducibility

The political application uses a corpus of California State Assembly member press releases (2015–2025), partitioned by the issuing member’s party, of which the AI-related subset is analysed here; a parallel K–12 education-policy corpus is used to test domain transfer. All code, the benchmark harness, the LLM-judge labels, the retrieved references, and the figure scripts are released at <https://github.com/bdesmarais/ai-policy-science-use>; every number is reproducible from committed outputs with open models and no paid API.

Data and code availability

All code, the benchmark-validation harness, the LLM-judge labels, the retrieved references, and the figure scripts are available at <https://github.com/bdesmarais/ai-policy-science-use>. Every result is reproducible from committed outputs using open-source models with no paid API. The validation benchmarks are SciFact (Wadden et al., 2020) and Climate-FEVER (Diggelmann et al., 2020); reference metadata are retrieved from OpenAlex (Priem et al., 2022).

AI-assistance disclosure

By design, the measurement instrument in this paper uses LLMs as components, and we disclose this explicitly. Stance “gold” labels were produced by Claude Opus 4.8 acting as an in-context judge (no fine-tuning, no API key); scalable surrogate labels were produced by open-source NLI models; an open-source panel (Qwen2.5, Phi-3.5, OLMo-2) provided an independent cross-developer check; and the same high-capacity model assisted in drafting and in executing the analysis. Every cited reference was checked against a primary source. The validation against human-labeled benchmarks is what licenses these uses; all design decisions and claims are the authors’. Earlier pipeline development was supported by NSF awards 2318460 and 2148215.

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